

Airbus Product Standard
Nutplate, rivetless,
Floating, variable counterbore,
Replaceable nut element

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1 Scope

This standard specifies the configuration, dimensions, tolerances and mass of a rivetless anchor nut for the realization of threaded and removable structural joints as an alternative for riveted anchor nuts.

Note: The application of rivetless nutplates per ABS1295 is limited to metallic structures as specified within ABD0090 and AIPS01-02-016.

Rivetless nutplates per ABS1295 with material codes "K" and "V" (see Table 1) shall be used for Aluminium structures with in-service temperatures from -60°C (-76°F) up to 190°C (375°F). For titanium applications up to 232°C (450°F), material code "N" can be selected. For high temperature titanium applications up to 426°C (800°F), rivetless nutplates per ABS1295 with material code "P" can be selected.

2 Normative References

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ABD0090	Fastener Selection and Joint Requirements for New Generation Composite Aircraft
ABS2120	Nut, self-locking, for use on rivetless nutplates, copper free
AIPS01-02-016	Installation of rivetless nutplate
AMS2700	Passivation treatments for corrosion resistant steel ¹
AMS5525	Steel, corrosion and heat resistant, sheet, strip, and plate, 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V, 1800 °F (982 °C) solution heat treated ¹
AMS5731	Steel, corrosion and heat-resistant, bars, wire, forgings, tubing, and rings, 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V consumable electrode melted, 1800 °F (982 °C) solution heat treated ¹
AMS5732	Steel, corrosion and heat resistant, bars, wire, forgings, tubing, and rings, 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V consumable electrode melted 1800 °F (982 °C) solution and precipitation heat treated ¹
AMS5737	Steel, corrosion and heat-resistant, bars, wire, forgings, and tubing, 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V consumable electrode melted 1650 °F (899 °C) solution and precipitation heat treated ¹
AS8879	Screw threads – UNJ profile, inch-controlled radius root with increased minor diameter ¹
ASTM A313	Standard specification for stainless steel spring wire ²
EN2424	Aerospace series – Marking of aerospace products
ISO2768-1	General tolerances – Tolerances for linear and angular dimensions without individual tolerance indications
MIL-C-5541	Chemical conversion coatings on aluminium and aluminium alloys ³
MIL-DTL-83488	Coating, Aluminium, High Purity ³

¹ Published by: Society of Automotive Engineers (SAE), 400 Commonwealth Drive, Warrendale, PA 15096-0001, USA

² Published by: American Society for Testing and Materials (ASTM),
100 Barr Harbor Drive, West Conshohocken, PA 19428-2959, USA

³ Published by: Department of Defense (DoD), the Pentagon, Washington, D.C. 20301, USA

NAS1794	Nut, self-locking, counter bore replaceable element ⁴
NAS1795	Retaining clip – Replaceable nut plate ⁴
NAS4006	Aluminum coating ⁴
NASM25027	Nut, self-locking, 250°F, 450°F and 800°F ³

3 Requirements

3.1 Configuration, Dimensions, Tolerances and Mass

3.1.1 The configuration, dimensions, tolerances and masses shall comply with Figures 1 and 2 and Tables 2 and 3 (Nominal) and Tables 4 and 5 (Oversize).

3.1.2 Grip lengths shall be in accordance with Table 6.

3.1.3 Tolerances not specified shall be in accordance with ISO2768-m.

3.1.4 The rivetless nutplate per ABS1295 is available in two variants:

- **Variant A** with regular floating nut element (not less than 0,76 mm absolute floating for –08 size; not less than 1,52 mm floating absolute for –3 and –4 size); nut element shall be according to ABS2120A. The retaining clip shall be according to NAS1795.
- **Variant B** with large floating nut element (not less than 3 mm floating absolute); nut element shall be according to ABS2120B. Retaining clip per manufacturer's option.

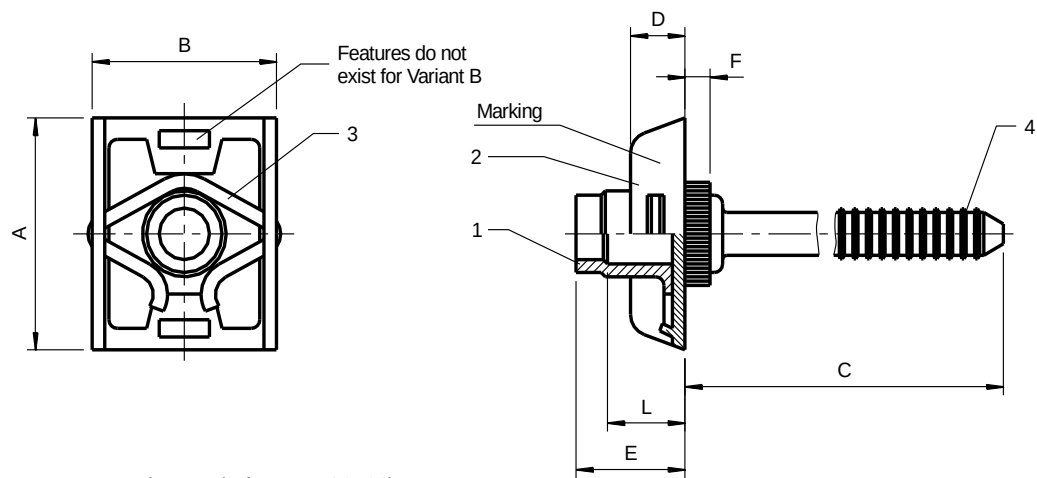
3.2 Material and Surface Treatment

The material and surface treatment shall be in accordance with Table 1.

⁴ Published by: Aerospace Industries Association of America, Inc. (AIA),
1250 Eye Street, N.W., Washington, D.C. 20005-3924, USA

Table 1: Material and Surface Treatment

Material code	Item						
	1	2		3		4	
	Nut element	Retainer		Retaining Clip		Mandrel	
	Reference	Material	Finish	Material	Finish	Material	Finish
D ⁴⁾	For variant A: ABS2120A*-*V; For variant B: ABS2120B*-*V	CRES A286 per AMS5525	Aluminium coating per NAS4006 NC	CRES XM-28 per ASTM A313 or A286 per AMS5525	Passivated per AMS2700	CRES A286 per AMS5731	Passivated per AMS2700 ⁵⁾
K ¹⁾	For variant A: ABS2120A*-*; For variant B: ABS2120B*-*		Passivated per AMS2700 and Aluminium pigmented coating and corrosion resistant Aluminium coat in accordance with NAS4006				
N ²⁾							
P ³⁾	For variant A: ABS2120A*-*P; For variant B: ABS2120B*-*P		Passivated per AMS2700				
V ⁴⁾	For variant A: ABS2120A*-*V; For variant B: ABS2120B*-*V		IVD coating per MIL-DTL- 83488, class 2, type II; yellow chromated per MIL-C- 5541 class 1A and class 3				
¹⁾ For aluminium structure applications ²⁾ For titanium structure applications up to 450°F (232°C) ³⁾ For titanium structure applications up to 800°F (426°C) ⁴⁾ Configuration shall only be used if electrical conductivity is required ⁵⁾ Lubrication of mandrel per manufacturer's option (copper not permitted)							



Caption:

- 1 – Nut element (refer to ABS2120)
- 2 – Nutplate (Retainer)
- 3 – Retaining Clip
- 4 - Mandrel

Figure 1: Nutplate configuration and dimensions (schematic)

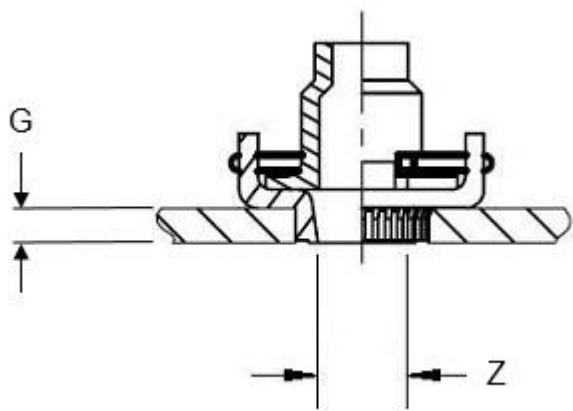


Figure 2: Installed nutplate (mandrel disposed after installation)

3.3 Dimensions, tolerances and mass – Nominal diameter

Table 2: Dimensions, tolerances and mass for variant A nutplates

Dimensions in mm (Inch)

Diameter code	Nut element per ABS2120A			A	B	C		D	E	Z	Mass ²⁾
	Thread size	C'bore dash no.	L Min.			Max.	Max.				Ref.
				(Stand)	(Short) ₁₎			Ref.	Max.	Min.	
08	.1640- 32 UNJC- 3B	-2	3,43 (.135)	15,11 (.595)	12,45 (.490)	36,83 (1.450)	27,94 (1.100)	6,22 (.245)	8,13 (.320)	5,54 (.218)	3,08 (6.8)
		-3	5,03 (.198)						9,73 (.383)		3,31 (7.3)
		-4	6,60 (.260)						11,33 (.446)		3,54 (7.8)
		-5	8,20 (.323)						12,90 (.508)		3,76 (8.3)
3	.1900- 32 UNJF- 3B	-1	1,93 (.076)	15,11 (.595)	12,45 (.490)	36,83 (1.450)	27,94 (1.100)	6,22 (.245)	6,55 (.258)	6,30 (.248)	2,75 (6.1)
		-2	3,43 (.135)						8,13 (.320)		3,08 (6.8)
		-3	5,03 (.198)						9,73 (.383)		3,31 (7.3)
		-4	6,60 (.260)						11,33 (.446)		3,54 (7.8)
		-5	8,20 (.323)						12,90 (.508)		3,76 (8.3)
4	.2500- 28 UNJF- 3B	-1	1,80 (.071)	18,80 (.740)	15,11 (.595)	34,93 (1.375)	27,94 (1.100)	6,73 (.265)	7,70 (.303)	7,87 (.310)	4,63 (10.2)
		-2	3,43 (.135)						9,27 (.365)		5,08 (11.2)
		-3	5,03 (.198)						10,83 (.427)		5,49 (12.1)
		-4	6,60 (.260)						12,42 (.489)		5,85 (12.9)
		-5	8,20 (.323)						14,00 (.551)		6,30 (13.9)
¹⁾ Only applicable for grip codes 01 and 02 ²⁾ Installed total weight (based on 02 grip code for diameter code 08 & 3 and 03 grip code for diameter code 4 respectively)											

Table 3: Dimensions, tolerances and mass for **variant B** nutplates

Dimensions in mm (Inch)

Diameter code	Nut element Per ABS2120B			A	B	C		D	E	Z	Mass ²⁾				
	Thread size	C'bore dash no.	L Min.			Max.	Max.				Ref.	Ref.	Max.	Min.	Ref.
															Kg / 1000 pcs
3	.1900-32 UNJF-3B	-2	3,25 (.128)	17,78 (.700)	15,24 (.600)	35,56 (1.400)	28,70 (1.130)	8,13 (.320)	8,76 (.345)	7,87 (.310)	5,94 (13.10)				
		-3	4,85 (.191)						10,36 (.408)		6,05 (13.34)				
		-4	6,43 (.253)						11,96 (.471)		6,31 (13.91)				
		-5	8,03 (.316)						13,54 (.533)		6,57 (14.48)				
4	.2500-28 UNJF-3B	-2	3,38 (.133)	18,16 (.715)	17,91 (.705)	35,56 (1.400)	28,70 (1.130)	9,91 (.390)	10,03 (.395)	9,47 (.373)	9,02 (19.89)				
		-3	4,98 (.196)						11,61 (.457)		9,17 (20.22)				
		-4	6,55 (.258)						13,18 (.519)		9,39 (20.70)				
		-5	8,15 (.321)						14,76 (.581)		9,75 (21.50)				
¹⁾ Only applicable for grip code 02															
²⁾ Installed total weight (based on 02 grip code for diameter code 08 and 3 and 03 grip code for diameter code 4 respectively)															

3.4 Dimensions, tolerances and mass - Repair sizes

Table 4: Dimensions, tolerances and mass for variant A nutplates oversize “X” for repair purposes
Dimensions in mm (Inch)

Diameter code	Nut element per ABS2120A			A	B	C		D	E	Z	Mass ²⁾
	Thread size	C'bore dash no.	L Min.			Max.	Max.				(Stand)
				Ref.	Ref.			Max.	Min.	Kg / 1000 pcs (lb / 1000 pcs)	
08	.1640-32 UNJF-3B	-2	3,43 (.135)	15,11 (.595)	12,45 (.490)	36,83 (1.450)	27,94 (1.100)	6,22 (.245)	8,13 (.320)	6,30 (.248)	2,95 (6.5)
		-3	5,03 (.198)						9,73 (.383)		3,18 (7.0)
		-4	6,60 (.260)						11,33 (.446)		3,40 (7.5)
		-5	8,20 (.323)						12,90 (.508)		3,63 (8.0)
3	.1900-32 UNJF-3B	-1	1,93 (.076)	15,11 (.595)	12,70 (.500)	36,83 (1.450)	27,94 (1.100)	6,22 (.245)	6,68 (.263)	6,30 (.248)	2,75 (6.1)
		-2	3,43 (.135)						8,26 (.325)		3,08 (6.8)
		-3	5,03 (.198)						9,86 (.388)		3,31 (7.3)
		-4	6,60 (.260)						11,45 (.451)		3,54 (7.8)
		-5	8,20 (.323)						13,03 (.513)		3,76 (8.3)
4	.2500-28 UNJF-3B	-1	1,80 (.071)	18,80 (.740)	15,24 (.600)	34,93 (1.375)	N/A	6,73 (.265)	7,82 (.308)	7,87 (.310)	4,63 (10.2)
		-2	3,43 (.135)						9,40 (.370)		5,08 (11.2)
		-3	5,03 (.198)						10,97 (.432)		5,49 (12.1)
		-4	6,60 (.260)						12,55 (.494)		5,85 (12.9)
		-5	8,20 (.323)						14,12 (.556)		6,30 (13.9)
¹⁾ Only applicable for grip codes 01 and 02											
²⁾ Installed total weight (based on 02 grip code for diameter code 08 and 3 and 03 grip code for diameter code 4 respectively)											

Table 5: Dimensions, tolerances and mass for variant B nutplates oversize “X” for repair purposes
Dimensions in mm (Inch)

Dimensions in mm (inch)

Diameter code	Nut element per ABS2120A			A	B	C		D	E	Z	Mass ²⁾
	Thread size	C'bore dash no.	L Min.			Max.	Max.				Ref.
3	.1900-32 UNJF-3B	-2	3,25 (.128)	17,78 (.700)	15,24 (.600)	35,56 (1.400)	28,70 (1.130)	8,13 (.320)	8,76 (.345)	7,87 (.310)	5,94 (13.10)
		-3	4,85 (.191)						10,36 (.408)		6,05 (13.34)
		-4	6,43 (.253)						11,96 (.471)		6,31 (13.91)
		-5	8,03 (.316)						13,54 (.533)		6,57 (14.48)

³⁾ Only applicable for grip codes 01 and 02

⁴⁾ Installed total weight (based on 02 grip code for diameter code 3)

3.5 Grip lengths

For the use of rivetless nutplates per ABS1295, it is mandatory to select the correct grip code depending from the material thickness of the concerned component. Available grip lengths can be selected from Table 6.

Table 6: Grip lengths

Grip code ¹⁾	Variant A			Variant B		
	G		F	G		F
	Min.	Max. ⁴⁾	Ref.	Min.	Max. ⁴⁾	Ref.
01	1,02 (.040)	1,60 (.063)	0,81 (.032)	N/A	N/A	N/A
02 ²⁾	1,40 ³⁾ (.055)	3,20 (.126)	1,19 (.047)	1,40 (.055)	3,20 (.126)	1,19 (.047)
03	3,18 (.125)	6,00 (.236)	2,79 (.110)	N/A	N/A	N/A
¹⁾ Some grip codes are not available for certain diameters. Refer to Table 7 for available diameter/grip combinations. ²⁾ Marking on the retainer is “15” ³⁾ Minimum grip length is 1,6 mm for –08 thread size RNPs ⁴⁾ Maximum grip length specification only applies for “K” and “V” material codes						

3.6 Mechanical properties

The mechanical properties of the installed part shall meet the specifications of Table 7.

Notes: The mechanical properties given in Table 7 are applicable for rivetless nutplates per ABS1295 that are installed in accordance with AIPS01-02-016.

ABS1295 shall meet the requirements of NASM25027 regarding torque-out and push-out retention, except for the –08 diameter code and the 01 grip code of –3 diameter code with reduced values

Table 7: Mechanical properties

Diameter code	Grip code	Variant A		Variant B	
		Min. torque-out	Min. push-out load	Min. torque-out	Min. push-out load
		Nm (in.lbf)	N (lbf)	Nm (inlbf)	N (lbf)
08	01	4,51 (40)	333 (75)	N/A	
	02				
	03				
3	01	4,51 (40)	333 (75)	N/A	
	02	6,77 (60)	445 (100)	6,77 (60)	444 (100)
	03				
4	02	N/A		11,29 (100)	556 (125)
	03	11,29 (100)	556 (125)		

3.7 Installation

Rivetless nutplates per ABS1295 shall be installed in accordance with the specifications of AIPS01-02-016.

3.8 Floating

Rivetless nutplates per ABS1295 shall provide the absolute floating as specified in Table 8 as a minimum if installed in accordance with the specifications of chapter 3.7:

Table 8: Minimum absolute floating specifications

Grip code	Dimensions in mm (Inch)				
	Variant A, diameter code			Variant B, diameter code	
	08	3	4	3	4
01	0,76 (.030)	1,52 (.060)	1,52 (.060)	3,00 (.118)	3,00 (.118)
02					
03					

Nut element shall be capable of engagement with bolt in the maximum misaligned position.

4 Designation

This type of standard shall be designated according to the philosophy of the following example:

	Description block	Identity block							
	Rivetless nutplate	ABS1295	K	3	=	2	A	01	X
Number of this standard									
Material code (see Table 1)									
Diameter code (see Table 2, Table 3, Table 4, Table 5)									
Mandrel length (see Table 2, Table 3, Table 4, Table 5) "–" = Standard "S" = Short									
Nut counterbore dash code (see Table 2, Table 3, Table 4, Table 5)									
Variant A or B (see Table 2, Table 3, Table 4, Table 5)									
Nut plate grip code (see Table 6)									
Oversize Code "None" = Regular size (see Table 2, Table 3) "X" = Oversize (see Table 4, Table 5)									

5 Marking

EN2424, style F and according to the specifications as shown in Figure 3.

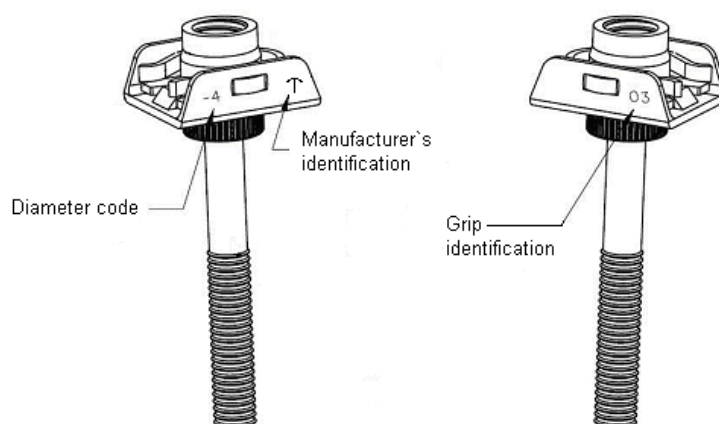


Figure 3: Retainer marking of Rivetless nutplates per ABS1295.

6 Technical Specification

Qualification and acceptance testing shall follow the specifications of NASM25027 but with the following amendments:

- Minimum mechanical torque- and push-out retention forces shall be in accordance with the specifications of Table 7.
- All mechanical requirements (e.g. torque-out and push-out load) as specified herein shall be met by the system if installed in accordance with AIPS01-02-016 - but without sealant and without any consecutive surface protection re-treatments - into metallic blank panels made from Al 2024-T351 or equivalent strength Aluminium material with the minimum thickness as specified per Table 6 in an installation hole of the maximum diameter as permitted by the AIPS with a tolerance of $+0/-0,02\text{mm}$.
- Visual examination as specified per NASM25027 shall be conducted with both uninstalled and installed retainers. For visual examination of installed retainers, each six retainers shall be installed in accordance with AIPS01-02-016 - but without sealant and without any consecutive surface protection re-treatments - into metallic blank panels made from Al 2024-T351 or equivalent strength Aluminium material with the minimum thickness as specified per Table 6 in an installation hole of the minimum diameter as permitted by the AIPS with a tolerance of $-0/+0,02\text{mm}$. Any evidence of cracking of the flange as a result of the installation shall cause the rejection of the lot.
- Oversize parts ("X") for repair purposes shall meet the mechanical requirements after being installed into holes that were reamed to the applicable maximum hole diameter as specified for the repair size by AIPS01-02-016 with a tolerance of $+0/-0,02\text{mm}$. These holes shall be prepared by reaming of existing holes in which a regular-sized RNP per ABS1295 of the same thread diameter was installed and then removed in accordance with AIPS01-02-016.

7 Example of Installation

The installed nut plate is shown in Figure 4.

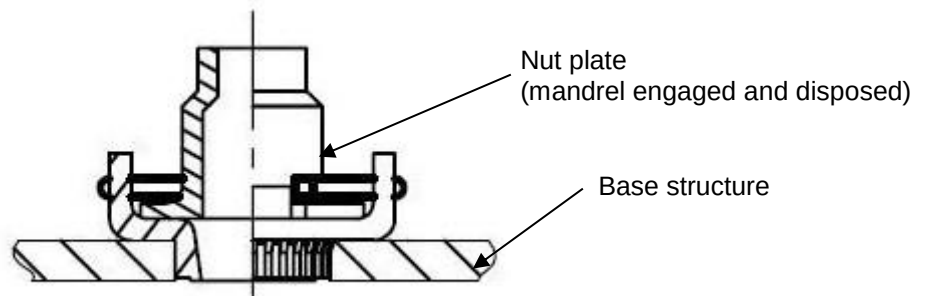


Figure 4: Example of installation

8 Record of Revisions

Issue	Clause modified	Description of modification
1 10/05	All	New standard
2 05/06	Clause 2	EN2000 replaced by AS9100
	Table 2	Dimension "L _{min} " changed for diameter code 4, counterbore dash no. -1 (was: 1,93mm / is: 1,80 mm)
	Tables 2, 3 and 4	UNJC changed in UNJF for diameter code 3 Dimension "C (Standard)" and "C (Short)" added Correction of dimensions "E" and "N _{Max} " Note "Retaining clip to be at manufacturer's discretion" changed into "Retaining clip at manufacturer's option"
	Clause 3.1.3	Note "Face side and chamfer manufacturer's option" changed into "Coating on grooves and chamfer at manufacturer's option"
	Table 1	Corrected in accordance with manufacturing drawings
	Figure 3	Designation table changed
	Clause 4	
3 07/07	All	Editorial revision
	Clause 1	Note added: "The application of rivetless nutplates variant B (large float) is limited
	Clause 2	to military programs".
	Table 1	AS9100 deleted
	Table 2	Note 2 added
	Table 3	Note 3 added
	Tables 4 and 5	Note added: "For military programs only!"
	Table 6	Added
	Table 7	Note added: "For military programs only!"
	Clause 3.6	Note 3 added
	Clause 4	Added
		Designation corrected (was ABS1295-K... / is: ABS1295K...)
	Clause 5	Oversize code "X" added
	Clause 6 and 7	Marking of diameter code optional AIPS01-02-016 added

4 02/10	All	Remark "For military programs only!" removed
	All	ABS2120 added as reference for nut element
	Clause 1	Sentence removed: "The application of rivetless nutplates variant B (large float) is limited to military programs." Sentence amended: "The application of rivetless nutplates is limited to aluminium structure (2024-T3, 7075-T73 or equivalent strength)." to "The application of rivetless nutplates is limited to metallic structures as specified within the limitations of the applicable process specification AIPS01-02-016."
	Clause 2	MIL-C-5541 and MIL-DTL-83488 added
	Clause 3	Variant description added, mandrel disposure remark deleted, figures added.
	Clause 3.1.1	Floating requirement added, variants added
	Clause 3.1.2	Chapter 3.1.2 added: Dimensions, tolerance and mass – regular size Regular size tables added Nut dimension table added 08 size added
	Clause 3.1.3	Chapter 3.1.2 added: Dimensions, tolerance and mass – Repair sizes Oversize tables added
	Clause 3.2	Material table extended for "T" and "V" configuration; Notes 3 to 6 added, table transposed
	Clause 3.3	Chapter 3.3 "Floatability" deleted → new chapter 3.3 is former chapter 3.4 "grip length"; Max. grip length of -03 grip code increased to 6mm
	Clause 3.3	Headline renamed to "Grip length"; note added that max grip length is only applicable for "K" and "V" material codes
	Clause 3.4	Sentence amended: "The mechanical properties given in table 8 are applicable for ABS1295 installed into the hole sizes and materials specified in AIPS01-02-016. ABS1295 meets the requirements of NASM25027 for torque-out and push-out load, except for diameter code 3 with 01 grip code with lower values." Changed to "The mechanical properties given in table 8 are applicable for rivetless nutplates per ABS1295 that are installed in accordance with the specifications of AIPS01-02-016. ABS1295 shall meet the requirements of NASM25027 regarding torque-out and push-out retention, except for the 01 grip code with reduced values."
	Clause 4	
	Clause 6	Designation extended: Plating and size code added for 800°F nuts Technical specification reworded Deviations from generic spec NASM25027 incorporated Mechanical properties linked to table 7 Visual examination requirement of installed retainers added Requirements on hole preparation for repair sizes added

Issue	Clause modified	Description of modification
	Table 4,5,8	Table layout revised.
09/22	Table 5	Corrosion resistant Aluminium coat per NAS4006 added to Retainer finish.
5 03/23	Table 1	Material Code D added for A220 program CPPR24424.